PHYSICS

1. B) $[ML^2T^{-2}]$ | Explanation : Torque ( ) = Force Perpendicular Distance. $[M L T^{-2}]\times[L]=[M L^2 T^{-2}]$.
2. B) N/m | Explanation : Surface tension is defined as force per unit length acting on the surface of a liquid. Hence, its unit is N/m.
3. C) The net external torque acting on it is zero. | Explanation : From Newton's second law for rotation, τ_{ext} =dL/dt. If net external torque τ_{ext} is zero, then dL/dt=0, meaning angular momentum (L) is conserved.
4. C) Zero | Explanation : Both the man and the coin are in a state of free fall, accelerating downwards at exactly g. Their relative acceleration (g-g) is zero.
5. B) 1/2 MR^2 | Explanation : The moment of inertia of a uniform solid cylinder about its geometric axis is mathematically derived as 1/2 MR^2
6. B) 100 m | Explanation : Displacement is the area under the v-t graph. Area of the triangle = 1/2 bh =1/2 10 × 20 = 100. m.
7. B) 4 m/s^2 | Explanation : Initial velocity u = 72 km/h = 72 \times (5/18) = 20 m/s. Final velocity v=0. Time t=5 s. Retardation a=(u-v)/t=20/5= 4 m/s^2
8. C) Parabola | Explanation : A projectile under uniform gravity with no air resistance follows a strictly parabolic path dictated by the equation y=x$tan\theta$ - (gx^2})/($2u^2\cos^2\theta$).
9. D) m^0 (Independent of mass) | Explanation : Escape velocity $v_e=\sqrt{\frac{2GM}{R}}$ It depends solely on the mass (M) and radius (R) of the planet, and is completely independent of the object's mass.
10. A) $\sqrt{gR}$ | Explanation : Orbital velocity $v_0=\sqrt{\frac{GM}{R}}$ Since $g=GM/R^2$ substituting $GM=gR^2$ gives $v_0=\sqrt{gR}$.
11. C) Temperature | Explanation : The zeroth law states that if two systems are in thermal equilibrium with a third, they are in equilibrium with each other. This defines the concept of temperature as the common property.
12. A) -40^0 | Explanation : Using the formula C/5=(F-32)/9. Letting C=F=x, x/5=(x-32)/9 -> 9x = 5x - 160 -> 4x = -160 -> x = -40^0
13. D) Zero | Explanation : Isochoric means constant volume. Since Work (W) = PΔV, and \Delta V=0 the work done is strictly zero.
14. B) Absorbs all radiations incident on it, regardless of wavelength. | Explanation : A perfectly black body absorbs 100% of the electromagnetic radiation falling on it, irrespective of the frequency or angle of incidence.
15. B) 40 J | Explanation : The work done in a cyclic process equals the enclosed area of the P-V loop. Area = (30-10) \times (3-1) = 20 N/m^2 × 2 m^3 = 40J.
16. A) 25% | Explanation : Efficiency = 1 - (T_{sink}/T_{source}). T_{source} = 127 + 273 = 400 K, Ts_{ink} = 27 + 273 = 300 K, \eta = 1 - (300/400) = 1 - 0.75 = 0.25 or 25%.
17. C) Radiation | Explanation : Radiation transfers heat via electromagnetic waves, which easily propagate through a perfect vacuum.
18. B) Diffraction | Explanation : Diffraction is the fundamental wave property where waves bend around the edges of an obstacle or slit of size comparable to their wavelength.
19. D) Violet | Explanation : According to Cauchy's equation, refractive index increases as wavelength decreases. Violet has the shortest visible wavelength, hence the maximum refractive index.
20. A) +33.3 cm | Explanation : Power of combination P = P_1 + P_2 = +5D - 2D = +3 D. Focal length f = 1/P = 1/3 meters = +33.3 cm.
21. B) It forms a virtual, erect, and diminished image, giving a much wider field of view. | Explanation : Convex mirrors always form erect, diminished images, which allows a driver to compress a very wide rear field of view into a small mirror.
22. C) The wavelength of the light used | Explanation : Fringe width $\beta = \lambda$ D/d. It is directly proportional to the wavelength (λ) and inversely proportional to the slit spacing (d).
23. B) 2.0 | Explanation : Refractive index n = 1/sin(C). If C = 30^0, then sin(30^0) = 0.5. Hence, n = 1/0.5 = 2.0
24. B) 1+ (D/f) | Explanation : For a simple microscope with the image at the near point, the angular magnification is given by M = 1 + D/f
25. B) Concave lens | Explanation : In myopia, the image converges in front of the retina. A diverging (concave) lens spreads the light rays before they enter the eye, pushing the focal point back onto the retina.
26. C) Conductivity | Explanation : Conductivity ( ) is mathematically defined as the reciprocal of electrical resistivity (p). $\sigma = 1/\rho$.
27. B) Right-Hand Thumb Rule | Explanation : The Right-Hand Thumb Rule dictates that if the thumb points in the current's direction, the curled fingers indicate the magnetic field's direction.
28. B) R/3 | Explanation : For n identical resistors in parallel, R_{eq} = R/n. Here n = 3, so R_{eq} = R/3.
29. B) 10 Ω | Explanation : In a balanced Wheatstone bridge (P/Q=R/S) no current flows through the central galvanometer arm. The equivalent resistance of identical arms R is simply R (10 Ω).
30. D) 600 C | Explanation : Charge Q = It. Time = 2 min = 120 s. Q = 5 A × 120 s = 600 Coulombs.
31. B) 0.1 N | Explanation |Force F = BIL sin(θ). Since it's perpendicular, θ = 90^0 and sin(90^0) = 1. F = 0.1 T × 2A × 0.5 m = 0.1 N
32. C) Eddy current losses | Explanation : Laminating the core with thin insulating layers drastically restricts the circular paths available for induced eddy currents, minimizing associated heat loss.
33. B) A m^2 | Explanation : Magnetic dipole moment (M) = Current × Area. The unit is Amperes \times square meters (A m^2).

34. C) Diamagnetic | Explanation : Diamagnetic materials align opposite to external fields, causing a weak repulsive force.
35. A) Zero | Explanation : Charges reside entirely on the outer surface of a conductor. Inside, the electric field is strictly zero, providing electrostatic shielding.
36. C) One-fourth | Explanation : Coulomb's Force $F \propto 1/r^2$. If the distance (r) is doubled (2r), the force becomes $1/(2r)^2$ = 1/4th of the original force.
37. B) 1.33 μF | Explanation : For capacitors in series, $1/C_{eq} = 1/C_1 + 1/C_2$. $1/C_{eq} = 1/2 + 1/4 = 3/4$. $C_{eq} = 4/3 = 1.33\ \mu$F.
38. C) 1/2 CV^2 | Explanation : The electrical potential energy stored in the electric field of a capacitor is mathematically given by $E = 1/2\ CV^2$.
39. C) Photon | Explanation : Photons are the gauge bosons of the electromagnetic force. They travel at the speed of light, with zero rest mass and zero charge.
40. C) Both electric and magnetic fields | Explanation : Because cathode rays are composed of negatively charged electrons, they experience Lorentz forces from both electric and magnetic fields.
41. C) NAND Gate | Explanation : The symbol shows an AND gate followed immediately by an inversion bubble (NOT), forming a NAND gate (Not AND).
42. B) Holes | Explanation : In n-type materials (doped with pentavalent impurities), electrons are the majority carriers, leaving holes as the minority carriers.
43. C) 1/8 | Explanation : Number of half-lives (n) = 15 / 5 = 3. The fraction remaining = $(1/2)^n = (1/2)^3 = 1/8$.
44. B) Balmer series | Explanation : The Balmer series involves transitions falling back to the n=2 shell, releasing photons precisely in the visible light spectrum.
45. A) The same number of protons but different number of neutrons. | Explanation : Isotopes occupy the same position in the periodic table (same protons/atomic number) but have different masses due to varying neutron counts.
46. B) Decreases | Explanation : Energy of state n is proportional to $-1/n^2$. As n gets larger, the shells become much closer together in energy.
47. B) 6200 Å | Explanation : Threshold wavelength λ_0 = hc / Work Function = 12400 eV·Å / 2.0 eV = 6200 Å.
48. C) Thermionic emission | Explanation : Thermionic emission provides thermal energy to overcome the surface work function, liberating electrons.
49. C) 100 Hz | Explanation : A full-wave rectifier flips the negative half-cycles of the AC wave to positive, creating two pulses per input cycle. Output frequency = 2×50 = 100 Hz.
50. D) Energy | Explanation : The equation states that the energy of the incident photon ($h\nu$) equals the binding energy overcome (Φ) plus the remaining kinetic energy, perfectly conserving energy.

CHEMISTRY

51. A) 0.529 Å | Explanation : The Bohr radius (a0) for the innermost orbit (n=1) of a hydrogen atom is defined as 0.529 Å (or 52.9 pm).
52. B) Square root of its molar mass | Explanation : Graham's Law states that the rate of effusion/diffusion of a gas is inversely proportional to the square root of its molar mass (or density) at constant temperature and pressure.
53. B) First law of thermodynamics (Conservation of Energy) | Explanation : Hess's Law states that the total enthalpy change for a reaction is the same regardless of the path taken. This directly stems from the First Law of Thermodynamics (Energy conservation).
54. B) First order | Explanation : All natural radioactive decay processes occur at a rate proportional to the current amount of undecayed substance, which defines first-order kinetics.
55. B) m = Z I t | Explanation : Faraday's First Law relates mass deposited (m) to the charge (Q=It). Thus, m=ZIt, where Z is the electrochemical equivalent.
56. B) 6 | Explanation : In the face-centered cubic (fcc) lattice of NaCl, each Na^+ is octahedrally surrounded by 6 Cl ions, and vice versa.
57. A) Increases with an increase in temperature | Explanation : Henry's constant KH is a function of the nature of the gas and increases with an increase in temperature, implying lower solubility of gases in liquids at higher temperatures.
58. C) m^{-1} | Explanation : Cell constant (G^0) = length / Area (l/A). Its unit is $m/m^2 = m^{-1}$ or cm^{-1}.
59. C) Heisenberg's Uncertainty Principle | Explanation : Heisenberg's Uncertainty Principle dictates that it is impossible to simultaneously determine the exact position and momentum of an electron, ruling out strict Bohr 'orbits'.
60. C) Greater than zero | Explanation : By the Second Law of Thermodynamics, any spontaneous process occurring in an isolated system (universe) leads to an increase in total entropy ($\Delta S > 0$).
61. B) A-B interactions are weaker than A-A and B-B interactions. | Explanation : Positive deviation occurs when solute-solvent (A-B) attractive forces are weaker than pure component (A-A or B-B) forces, allowing easier vaporization and higher vapor pressure.
62. B) H_2O forms extensive intermolecular hydrogen bonds. | Explanation : Oxygen is highly electronegative and small in size, allowing water molecules to form a strong, extensive 3D network of hydrogen bonds, requiring immense energy to boil.

63. D) It is the negative electrode where oxidation occurs. | Explanation : In a galvanic (voltaic) cell, oxidation always occurs at the anode. Because it releases electrons into the external circuit, the anode acts as the negative terminal.
64. C) 36 g | Explanation : Stoichiometry: $2H_2$ (4g) + O_2 (32g) -> $2H_2O$. 32g of O_2 requires exactly 4g of H_2. There is no limiting reagent here. They form 36g of H_2O.
65. B) 3.5 | Explanation : For a weak acid, $[H^+] = \sqrt{(Ka * C)} = \sqrt{(10^{-5} * 0.01)} = \sqrt{(10^{-7})} = 10^{-3.5}$. Thus, pH = 3.5.
66. A) 0.01 s^{-1} | Explanation : For a 1st order reaction, k = 0.693 / t_1/2. k = 0.693 / 69.3 = 0.01 s^{-1}.
67. C) 1.04 V | Explanation : Nernst Eq: E = E° - (0.0591/n) * log($[Zn^{2+}]/[Cu^{2+}]$). E = 1.10 - (0.0591/2) * log(1/0.01) = 1.10 - 0.0295 * 2 = 1.10 - 0.059 = 1.04 V.
68. C) The extra stability of the exactly half-filled 2p orbital in Nitrogen | Explanation : Nitrogen has an electronic configuration of $1s^2\ 2s^2\ 2p^3$. The exactly half-filled 2p orbital provides extra exchange energy and stability, making it harder to remove an electron compared to Oxygen ($2p^4$).
69. C) sp^3d^2 | Explanation : Sulfur (Group 16) has 6 valence electrons and forms 6 sigma bonds with Fluorine. Steric number = 6. This corresponds to sp^3d^2 hybridization and octahedral geometry.
70. B) Chlorides and Sulfates of Calcium and Magnesium | Explanation : Temporary hardness is caused by bicarbonates. Permanent hardness, which cannot be removed by simple boiling, is caused by chlorides and sulfates of Ca and Mg.
71. C) Apple green | Explanation : In flame photometry, Barium salts consistently emit an apple green colored light due to distinctive electronic transitions.
72. B) Three-center two-electron bonds (Banana bonds) | Explanation : Diborane is electron-deficient. To satisfy stability, it forms two bridged B-H-B bonds, known as 3-center 2-electron (3c-2e) bonds or 'banana bonds.'
73. C) Group 11 | Explanation : Group 11 contains Cu, Ag, and Au, traditionally used to mint coins.
74. A) +2 | Explanation : Let Fe be x. Cyanide is -1. x + 6(-1) = -4. x = +2. Fe is in the +2 oxidation state.
75. B) Increasing nuclear charge coupled with very poor shielding by 4f electrons. | Explanation : As electrons fill the 4f orbitals across the lanthanide series, they provide very poor shielding against the increasing nuclear charge, causing a steady contraction in atomic radii.
76. C) In NF_3, the orbital dipole of the lone pair opposes the resultant N-F bond dipoles. | Explanation : In NH_3, the N-H bond dipoles reinforce the lone pair dipole. In NF_3, the highly electronegative Fluorine atoms pull the bond dipoles opposite to the lone pair dipole, reducing the net moment.
77. D) $HClO_4$ (Perchloric acid) | Explanation : Acid strength increases with the oxidation state of the central atom. In $HClO_4$ (+7 state), the 3 highly electronegative terminal oxygens efficiently delocalize the negative charge on the perchlorate anion, making it extremely stable.
78. A) Absorption of light facilitating d-d electron transitions within split d-orbitals. | Explanation : When ligands approach a metal ion, the degenerate d-orbitals split in energy. Visible light absorption excites an electron from a lower d-orbital to a higher one (d-d transition), transmitting the complementary color.
79. A) The ΔG vs T curve for M_1 is positioned below the curve for M_2 at that temperature. | Explanation : A metal can reduce the oxide of another metal if it has a greater affinity for oxygen. Thermodynamically, this occurs if its formation curve lies below the other metal's curve on the Ellingham Diagram (ΔG is more negative).
80. C) +1 | Explanation : In the dominant Lewis structure of Ozone (O=O+-O-), the central oxygen has 6 valence electrons, 3 bonds (one double, one single), and 1 lone pair. Formal Charge = 6 - (3+2) = +1.
81. C) 36 | Explanation : EAN = Z - (Oxidation State) + 2 * (Coordination Number). For Co in $[Co(NH_3)_6]^{3+}$: EAN = 27 - 3 + 2(6) = 24 + 12 = 36 (Krypton's noble gas configuration).
82. C) It attaches to the central metal ion through two or more donor atoms to form a ring. | Explanation : Chelation occurs when a single polydentate ligand binds to a central metal ion using multiple donor atoms, forming a highly stable ring structure (e.g., EDTA or ethylenediamine).
83. B) 3-Hydroxybutanoic acid | Explanation : Numbering starts from the carboxylic acid carbon. The hydroxyl group (-OH) is on the 3rd carbon of a 4-carbon chain. Thus, 3-Hydroxybutanoic acid.
84. B) 3 | Explanation : Pentane (C_5H_{12}) has three chain isomers: n-pentane, isopentane (2-methylbutane), and neopentane (2,2-dimethylpropane).
85. A) Symmetrical alkanes containing an even number of carbon atoms. | Explanation : Wurtz reaction joins two alkyl radicals. 2R-X + 2Na -> R-R + 2NaX. It effectively doubles the carbon chain, producing symmetrical alkanes.
86. C) Alkyl fluorides | Explanation : Swarts reaction uses heavy metal fluorides (like AgF, Hg_2F_2, CoF_2) to substitute chloride/bromide in alkyl halides, selectively yielding alkyl fluorides.
87. B) Concentrated HCl and Anhydrous $ZnCl_2$ | Explanation : Lucas reagent is an equimolar mixture of concentrated Hydrochloric Acid (HCl) and anhydrous Zinc Chloride (ZnCl2).
88. B) Ammoniacal silver nitrate solution | Explanation : Tollens' reagent is freshly prepared ammoniacal silver nitrate, containing the complex ion $[Ag(NH_3)_2]^+$, which is reduced to metallic silver by aldehydes.
89. A) Primary amines (aliphatic and aromatic) | Explanation : Only primary amines (aliphatic or aromatic) undergo the Carbylamine reaction when heated with Chloroform ($CHCl_3$) and alcoholic KOH to form vile-smelling isocyanides.
90. B) α-1,4 glycosidic linkage | Explanation : Maltose is a disaccharide formed by two α-D-glucose units joined by an α-(1->4) glycosidic linkage.

91. B) Isoprene (2-methyl-1,3-butadiene) | Explanation : Natural rubber is cis-1,4-polyisoprene, derived from the polymerization of isoprene (2-methyl-1,3-butadiene).
92. C) Complete inversion of configuration (Walden inversion) | Explanation : SN^2 mechanism involves a one-step backside attack by the nucleophile, causing the molecule to flip inside out, resulting in complete Walden inversion of stereochemical configuration.
93. B) $(CH_3)2NH > CH_3NH_2 > (CH_3)3N > NH_3$ | Explanation : In aqueous solution, basicity depends on +I effect, steric hindrance, and solvation by water. For methylamines, the observed order is strictly Secondary > Primary > Tertiary > Ammonia.
94. B) (4n+2) π electrons | Explanation : Huckel's rule states that to be aromatic, a cyclic, planar, fully conjugated molecule must possess (4n + 2) π electrons, where n is an integer (0, 1, 2, ...).
95. C) The presence of at least one α-hydrogen atom. | Explanation : Aldol condensation utilizes the acidity of the α-hydrogen. Base removes this proton to form a nucleophilic enolate ion, which then attacks another carbonyl carbon.
96. B) It increases acidity by stabilizing the conjugate base (carboxylate ion) via the -I effect. | Explanation : Electron Withdrawing Groups (EWGs) pull electron density away from the carboxylate ion (-COO-). This disperses the negative charge, highly stabilizing the conjugate base, thereby enhancing acidity.
97. A) Due to the partial double bond character of the C-Cl bond arising from resonance. | Explanation : In chlorobenzene, the lone pairs on chlorine conjugate with the pi system of the benzene ring (+R effect). This resonance grants the C-Cl bond partial double bond character, making it very difficult to break.
98. D) Carboxylic acid | Explanation : Grignard reagent (R-MgX) acts as a nucleophile, attacking the electrophilic carbon of CO_2 (O=C=O) to form a carboxylate magnesium salt. Acid hydrolysis yields a carboxylic acid (R-COOH).
99. B) Two molecules of Acetaldehyde (Ethanal) | Explanation : 2-butene is CH_3-CH=CH-CH_3. Cleaving the double bond via reductive ozonolysis and adding an oxygen atom to each fragment yields two molecules of CH_3-CHO (Acetaldehyde).
100. B) 2-Butene | Explanation : Acid-catalyzed dehydration of 2-butanol forms a carbocation. According to Zaitsev's rule, the more highly substituted alkene is the major product due to hyperconjugation stability. Thus, 2-butene is the major product over 1-butene.

BOTANY

101. A) Carbohydrate | Explanation: carbs alpha napthol reacts with furfurals formed by the dehydration using h2so4 gives colored compound.
102. B) The glycosidic bond is formed between the anomeric carbon 1 of glucose and the anomeric carbon 2 of fructose, masking both reducing groups. | Explanation : A sugar is reducing if it has a free aldehyde or ketone group capable of acting as a reducing agent. In sucrose, the link joins the anomeric carbon of alpha-D-glucose (C_1) and the anomeric carbon of beta-D-fructose (C_2). Since both potential reducing centers are locked in the glycosidic bond, sucrose cannot open into a linear form to expose a free carbonyl group.
103. B) Lipopolysaccharide (LPS) layer of the outer membrane | Explanation : Gram-negative bacteria possess an outer membrane outside their thin peptidoglycan layer. This outer layer contains lipopolysaccharide (LPS) complexes. The Lipid A portion of LPS functions as a powerful endotoxin that triggers severe inflammatory cascades and septic shock in human patients.
104. B) Two adjacent cells of the exact same filament. | Explanation : Scalariform conjugation involves two parallel filaments aligning and forming conjugation tubes between opposing cells. Lateral conjugation occurs between two adjacent cells belonging to the same single filament, making it an example of homothallic self-fertilization.
105. B) Ascomycetes | Explanation : Ascomycetes (sac fungi) do not possess any motile flagellated stages. They form sexual spores called ascospores endogenously inside a specialized sac-like cell called an ascus. Basidiomycetes also lack motile cells but form sexual spores exologously on a basidium.
106. B) True unicellular elaters with spiral bands | Explanation : Marchantia capsules contain spores mixed with specialized elongated, sterile cells called elaters. These elaters have spiral cell wall thickenings that are highly hygroscopic. As they absorb or lose moisture, they twist twist vigorously, helping loosen and eject the spores.
107. C) Dictyostele | Explanation : A dictyostele is a type of siphonostele that is broken up into a ring of isolated vascular strands (meristeles) due to the presence of multiple overlapping leaf gaps, which is the characteristic anatomical configuration of the Dryopteris rhizome.
108. B) Sunken stomata placed deeply within pits, combined with a highly thick sclerenchymatous hypodermis | Explanation : Pinus needles have highly specialized xerophytic structures: a thick, waxy cuticle, a multi-layered sclerenchymatous hypodermis, sunken stomata tucked away in pits to reduce the transpiration rate, and a specialized transfusion tissue that aids lateral water movement.
109. C) Fabaceae | Explanation : These diagnostic floral traits strictly define the family Fabaceae (specifically the subfamily Papilionoideae). Features like a papilionaceous corolla, diadelphous stamens ((9)+1), and a single carpel with marginal placentation are textbook markers of this family.
110. B) The viral genome integrates cleanly into the host bacterial chromosome, replicating silently as a prophage without killing the host cell. | Explanation : In the lytic cycle, the phage injects its DNA, takes over the host machinery to make viral copies, and ruptures the host cell. In the lysogenic cycle, the viral DNA integrates directly into the host genome as a prophage, remaining dormant and copying itself along with the host cell division until triggered to enter the lytic phase.

111. B) Chronic systemic hypertension and neuropsychiatric agitation | Explanation : Reserpine, an alkaloid extracted from the roots of Rauwolfia serpentina, acts by permanently blocking the vesicular monoamine transporter (VMAT). This depletes peripheral catecholamines (like norepinephrine), leading to significant vasodilation and a steady drop in blood pressure, making it an effective antihypertensive and antipsychotic agent.
112. B) Nitrobacter | Explanation : Nitrification is a two-step chemoautotrophic pathway. First, bacteria like Nitrosomonas oxidize ammonia (NH^{4+}) into nitrites (NO^{2-}). Next, Nitrobacter species oxidize those nitrites into plant-absorbable nitrates (NO^{3-}).
113. C) Amphibious rooted plants like Typha, Phragmites, and Sagittaria | Explanation : The reed-swamp stage of a hydrosere occurs when the water body becomes shallow due to siltation. Rooted amphibious plants like Typha and Phragmites establish themselves, with their foliage projecting high above the water line, further accelerating soil accumulation.
114. B) UV radiation homolytically cleaves CFCs to release highly reactive free chlorine radicals, each capable of breaking down thousands of ozone (O_3) molecules into oxygen (O_2). | Explanation : When CFCs reach the stratosphere, intense ultraviolet light breaks them down, releasing free chlorine atoms (radicals). A single chlorine radical reacts with ozone (O3) to form chlorine monoxide (ClO) and O_2. The chlorine radical is then regenerated in a cyclical loop, allowing it to continually destroy tens of thousands of ozone molecules.
115. C) Betula utilis (Bhojpatra) and Rhododendron forest | Explanation : The sub-alpine zone of Nepal (3,000-3,800m) is characteristically dominated by Betula utilis (Himalayan Birch/Bhojpatra), famous for its smooth, white, papery bark that peels off in sheets, mixed alongside various high-altitude Rhododendron species.
116. C) Transverse 'flip-flop' movement of lipids from one leaflet to the opposing leaflet | Explanation : Lateral and rotational movements happen rapidly and spontaneously because they keep the non-polar tails in the hydrophobic interior. Transverse (flip-flop) movement requires a polar lipid head group to pass entirely through the non-polar core, making it rare unless catalyzed by specialized energy-dependent enzymes like flippases or floppases.
117. C) The presence of a circular, naked DNA genome coupled with prokaryotic-like 70S ribosomes. | Explanation : Both mitochondria and chloroplasts contain their own naked, circular DNA molecules (lacking histones) and 70S ribosomes, which are identical to those found in bacteria. This provides strong structural evidence that they evolved from free-living prokaryotes that were engulfed by ancestral eukaryotic cells.
118. C) Telomeres | Explanation : Telomeres are repetitive nucleotide sequences located at the termini of linear chromosomes. They act as protective caps that prevent chromosome ends from fusing or degrading. Due to the 'end-replication problem', telomeres shorten slightly with each cycle of cell division.
119. B) interphase in cell cycle
120. B) It binds to tubulin dimers, preventing spindle microtubule polymerization and arresting cells completely in Metaphase. | Explanation : Colchicine inhibits tubulin polymerization, preventing the formation of mitotic spindle fibers. Without spindle fibers, sister chromatids cannot align properly or separate during anaphase. This stalls the cell cycle at metaphase, when chromosomes are at their most condensed and visible state, making it ideal for karyotyping.
121. B) 10 base pairs | Explanation : B-DNA has a right-handed helical structure. A single full turn has a pitch of 3.4 nm (34 A) and contains exactly 10 base pairs, meaning the axial distance between adjacent base pairs is 0.34 nm (3.4 A).
122. B) lagging
123. C) Klinefelter's syndrome | Explanation : Klinefelter's syndrome is a sex-chromosome aneuploidy where a male individual inherits an extra X chromosome, resulting in an XXY configuration (total 47 chromosomes). Down's (Trisomy 21) and Edward's (Trisomy 18) are autosomal trisomies, while Turner's (45,XO) is a sex chromosome monosomy.
124. B) Syntenic and tightly linked together in the coupling (cis) configuration. | Explanation : A 1:1:1:1 ratio indicates independent assortment. A skewed ratio like 7:1:1:7 shows that the parental phenotypes (7+7) vastly outnumber the recombinant phenotypes (1+1). This is a hallmark of genetic linkage, where the two genes reside close together on the same chromosome, limiting independent assortment.
125. C) 18% | Explanation : According to Chargaff's Rules, in any double-stranded DNA molecule, % Adenine (A) = % Thymine (T), and % Guanine (G) = % Cytosine (C). If A = 32%, then T must also be 32%. Together, A + T = 64%. The remaining bases must be G + C = 100% - 64% = 36%. Because G = C, the percentage of Cytosine is 36% / 2 = 18%.
126. C) 2n = 21; highly sterile because the odd ploidy level prevents normal homologous chromosome pairing during Meiosis I | Explanation : The diploid parent produces haploid gametes (n = 7), and the tetraploid parent produces diploid gametes (2n = 14). Their fusion creates a triploid zygote (3n = 21). Triploid plants are viable and can grow normally via mitosis, but they are highly sterile because the three sets of chromosomes cannot pair evenly into homologous pairs during Zygotene/Pachytene of Meiosis I, leading to unbalanced, non-viable gametes.
127. B) Bicollateral bundle | Explanation : Bicollateral vascular bundles are characterized by a central strand of xylem flanked by phloem on both its outer (peripheral) and inner (pith) sides. They are always open, with cambium strips located between the xylem and both patches of phloem, typically seen in Cucurbitaceae.
128. C) A polyarch xylem configuration with more than six distinct xylem bundles, surrounding a large, well-developed central parenchymatous pith. | Explanation : Dicot roots typically have a small number of xylem bundles (2 to 6, i.e., diarch to hexarch) and a small or absent pith. Monocot roots have numerous xylem bundles (more than 6, i.e., polyarch) and a large, prominent, well-developed central pith.

129. B) Spring season; Spring wood (Early wood) | Explanation : In spring, environmental conditions favor active growth, and the plant requires efficient water transport. The cambium becomes highly active, producing xylem vessels with wide cavities and thin walls. This is spring wood (or early wood), which is lighter in density and color compared to dense, narrow autumn wood.
130. C) Zero | Explanation : By definition and international convention, the water potential of pure water at standard temperature and pressure is set to exactly zero. Adding solutes lowers this free energy state, making the solute potential (Ys) and resulting water potential (Yw) negative numbers.
131. B) ATP and NADPH + H^+ | Explanation : Non-cyclic photophosphorylation utilizes both Photosystem II and Photosystem I. The continuous unidirectional flow of electrons driven by light splitting of water generates a proton gradient that drives ATP synthesis, while terminal electron capture by $NADP^+$ reductase yields NADPH. Both compounds are required to power the dark reactions.
132. B) One molecule of 3-PGA and one molecule of 2-Phosphoglycolate | Explanation : When oxygen levels are high, Rubisco acts as an oxygenase. It adds O_2 to RuBP (5-carbon) to produce one molecule of 3-phosphoglycerate (3-carbon) and one molecule of 2-phosphoglycolate (2-carbon). Phosphoglycolate cannot be used directly in the Calvin cycle and must be recycled via a pathway spanning chloroplasts, peroxisomes, and mitochondria.
133. B) Succinyl-CoA into Succinate | Explanation : The cleavage of the high-energy thioester bond of Succinyl-CoA to form Succinate is catalyzed by the enzyme succinyl-CoA synthetase (succinate thiokinase). This reaction is coupled to the phosphorylation of GDP to GTP (or ADP to ATP in plants) via substrate-level phosphorylation.
134. B) DPD_X = 4 atm, DPD_Y = 8 atm, DPD_Z = 3 atm; water will move from Cell Z and Cell X into Cell Y. | Explanation : Diffusion Pressure Deficit (DPD) represents a cell's suction pressure and is calculated as DPD = OP - TP. For Cell X: DPD = 10 - 6 = 4 atm. For Cell Y: DPD = 12 - 4 = 8 atm. For Cell Z: DPD = 8 - 5 = 3 atm. Water always flows from an area of lower DPD (higher water potential) to an area of higher DPD (lower water potential). Because Cell Y has the highest suction pressure (8 atm), net water will flow from both Cell Z (3 atm) and Cell X (4 atm) into Cell Y.
135. B) Gibberellic acid (GA_3) | Explanation : Gibberellins (GA_3) are famously known for causing hyper-elongation of intact stem internodes (especially in dwarf varieties). They also play a core physiological role in breaking seed dormancy and inducing the synthesis of alpha-amylase enzymes in aleurone layers of germinating cereal grains.
136. C) 7 cells and 8 nuclei | Explanation : The standard Polygonum-type embryo sac contains 8 haploid nuclei distributed into 7 distinct cells: three antipodal cells at the chalazal end, one large central cell containing two polar nuclei, and a three-celled egg apparatus (one egg cell flanked by two synergids) at the micropylar end.
137. C) anatropous
138. B) Emasculation | Explanation : Emasculation is the process of removing stamens or anthers from bisexual female parent flowers before they mature and shed pollen. This step is essential in controlled cross hybridization to eliminate the risk of self-pollination.
139. B) T-DNA (Transfer DNA) region | Explanation : The Ti plasmid contains a specific segment called T-DNA flanked by 25-base-pair direct repeats. The virulence (vir) genes encode machinery that excises this single-stranded T-DNA fragment and transfers it directly into the host plant cell nucleus, where it integrates into the nuclear genome.
140. B) Polyethylene Glycol (PEG) and high Ca^{2+} concentration | Explanation : While cellulase and pectinase are used to digest cell walls to isolate naked protoplasts, the actual fusion of those isolated protoplast membranes requires a powerful fusogen. Polyethylene glycol (PEG) combined with alkaline high Ca^{2+} solutions alters membrane charges, facilitating successful protoplast aggregation and cell fusion.

ZOOLOGY

141. B) Chimpanzee | Explanation : At the molecular level, Chimpanzee Cytochrome c is entirely identical to human Cytochrome c, sharing 104 out of 104 amino acids in the exact same sequence. This biochemical homology provides definitive quantitative evidence of recent common ancestry compared to the Rhesus Monkey (which differs by 1 amino acid) or Yeast (which differs by dozens).
142. B) Somatoplasm variations do not influence the genetic architecture of germplasm cells within the gonads. | Explanation : Weismann's Germplasm Theory states that multicellular organisms consist of germ cells (which contain heritable information passed to offspring) and somatic cells (which carry out bodily functions). Changes or mutilations to somatic tissue (somatoplasm) do not alter the genetic information stored in the protected germline (germplasm), which is why acquired somatic changes are not inherited.
143. D) 0.0198 | Explanation : Let the frequency of the recessive allele be q, and the dominant allele be p. The frequency of the homozygous recessive phenotype is q^2 = 1/10,000 = 0.0001. Taking the square root, q = 0.01. Since p + q = 1, p = 1 - 0.01 = 0.99. Heterozygous carriers are represented by 2pq. Therefore, 2pq = 2 * 0.99 * 0.01 = 0.0198 (or roughly 1.98%).
144. B) Cnidocil | Explanation : The cnidocil is a hair-like, modified ciliated sensory process on the external surface of a cnidocyte. When physically touched by prey or stimulated chemically, it acts as a trigger mechanism, causing the operculum (lid) to fly open and explosively discharge the internal inverted venomous thread.
145. C) King Crab (Limulus) - Book gills | Explanation : Limulus (the King Crab or Horseshoe Crab), which is a 'living fossil', respires via flat, leaf-like abdominal appendages arranged like pages of a book, called book gills. Scorpions use book lungs, prawns use branchial gills, and centipedes use tracheal tubules.

146. B) A schizocoelom arises from the physical splitting of solid bilateral bands of embryonic mesoderm. | Explanation : In protostomes (Annelida, Arthropoda, Mollusca), the coelom is a schizocoelom, which develops when solid masses of mesoderm split internally to create the fluid-filled body cavity. In deuterostomes (Echinodermata, Chordata), it is an enterocoelom, formed by the out-pouching of the archenteron (primitive gut).
147. A) Balanoglossus (Hemichordata) | Explanation : The description matches Balanoglossus, an acorn worm belonging to Phylum Hemichordata. The anterior buccal diverticulum was historically mistaken for a primitive short notochord and is called a stomochord. Hemichordata are characterized by a tripartite body (proboscis, collar, trunk) and pharyngeal gill slits but lack a true chordate notochord.
148. B) Lining of the epididymis; composed of non-motile, elongated actin filament bundles | Explanation : Stereocilia are not true cilia; they lack microtubular 9+2 structures and axonemal dynein. They are structurally very long, branched, non-motile microvilli anchored by an actin filament core, found lining the epididymis and vas deferens where they increase surface area for fluid reabsorption.
149. B) One transverse (T) tubule flanked symmetrically by two terminal cisternae of the sarcoplasmic reticulum | Explanation : In skeletal muscle, a triad is a specific alignment where a single invagination of the sarcolemma, the T-tubule, is positioned between a pair of terminal cisternae (sac-like regions storing calcium) of the sarcoplasmic reticulum. This allows the action potential passing down the T-tubule to directly trigger calcium release.
150. B) The enzymatic hydroxylation of proline and lysine residues during procollagen synthesis | Explanation : Ascorbic acid is a required reducing cofactor for the enzymes prolyl hydroxylase and lysyl hydroxylase. These enzymes add hydroxyl groups (-OH) to proline and lysine amino acids within pre-procollagen chains. Hydroxyproline stabilizes the triple helix structure of collagen; without it, defective tropocollagen molecules cannot assemble properly, leading to scurvy.
151. B) Spatial summation will drive the membrane potential to -55 mV, triggering an action potential at the axon hillock. | Explanation : Total electrical change = (50 * +1.5 mV) + (30 * -2.0 mV) = +75 mV - 60 mV = +15 mV of net depolarization. Baseline potential = -70 mV. New potential = -70 mV + 15 mV = -55 mV. Since the potential reaches the exact threshold value of -55 mV, voltage-gated Na^+ channels at the axon hillock open and fire an action potential.
152. B) The Apical Complex (Rhoptries and Micronemes) | Explanation : The invasive stages of apicomplexan parasites (sporozoites and merozoites) possess a collection of specialized organelles at their anterior tip called the apical complex. This includes rhoptries and micronemes, which secrete ligands for cell binding and enzymes that distort the host cell membrane, allowing the parasite to enter.
153. B) 3rd segment | Explanation : In Pheretima, the nervous system begins with a pair of supra-pharyngeal (cerebral) ganglia fused together on the dorsal surface of the pharynx in the 3rd segment. They connect via circum-pharyngeal connectives in the 4th segment to the sub-pharyngeal ganglia.
154. C) An underdeveloped, vestigial ovary that can mature if the testes are surgically removed. | Explanation : Bidder's Organ is an embryonic remnant of an ovary found in male anurans (frogs and toads). If the testes are removed experimentally, Bidder's organ can develop into a functional, egg-producing ovary, demonstrating an underlying reproductive plasticity.
155. B) Intersegmental apertures of segments 5/6, 6/7, 7/8, and 8/9; they store foreign allogenic sperm received from a partner. | Explanation : Earthworms possess 4 pairs of spermathecae situated in segments 6, 7, 8, and 9, with external openings at the intersegmental furrows of 5/6, 6/7, 7/8, and 8/9. They receive and store spermatozoa from another earthworm during copulation, which are later utilized to fertilize eggs inside the cocoon.
156. B) It helps direct deoxygenated blood preferentially into the pulmocutaneous arches and oxygenated blood into the carotid and systemic arches. | Explanation : The spiral valve in the conus arteriosus of the frog heart divides the lumen dynamically during ventricular contraction. Because deoxygenated blood enters the conus first, the low-pressure path directs it primarily into the pulmocutaneous arch (to the lungs/skin). As pressure increases, the valve shifts to direct subsequent oxygenated blood into the systemic and carotid arches.
157. C) Calciferous glands of the esophagus | Explanation : The wall of the earthworm's esophagus contains specialized excretory structures called calciferous glands (in segments 10 to 14). They extract excess calcium and carbonate ions from the blood and secrete calcium carbonate crystals into the food. This neutralizes the humic acids present in ingested soil and leaf litter.
158. C) Sucrase - One molecule of Glucose + One molecule of Fructose | Explanation : Sucrase breaks down sucrose into glucose and fructose. Maltase cleaves maltose into two glucose molecules, and lactase breaks down lactose into glucose and galactose.
159. A) High oxygen partial pressure (pO_2) in the lungs promotes the unloading of carbon dioxide from hemoglobin. | Explanation : The Haldane Effect describes how oxygen binding to hemoglobin in the lungs alters its structure, making it less affine to carbon dioxide and H^+ ions, which promotes the release of CO_2 into alveolar spaces. (The opposite process—CO_2 promoting O_2 release is the Bohr Effect).
160. C) Ductus arteriosus | Explanation : The ductus arteriosus is a shunting blood vessel in the fetus that connects the pulmonary artery directly to the aorta, allowing right ventricular blood to bypass the uninflated lungs. It closes shortly after birth to form the fibrous ligamentum arteriosum.
161. C) Thick ascending limb of the Loop of Henle | Explanation : The thick ascending limb of the loop of Henle is completely impermeable to water because it lacks aquaporin water channels. However, it contains the $Na^+/K^+/2Cl^-$ co-transporter, which actively pumps solutes out of the tubular fluid, diluting the urine and establishing the medullary osmotic gradient.

162. B) Broca's Area | Explanation : Broca's Area, located in the posterior inferior frontal gyrus of the dominant hemisphere (usually left), is the motor speech area responsible for generating the muscular movements needed for articulate speech. Injury here causes Broca's (expressive) aphasia. Wernicke's area is in the temporal lobe and handles comprehension.
163. B) The photo-isomerization of 11-cis-retinal into all-trans-retinal | Explanation : Rhodopsin consists of the protein opsin bound to 11-cis-retinal. When struck by a photon, 11-cis-retinal instantly isomerizes into all-trans-retinal. This conformational shift triggers a G-protein pathway (transducin) that leads to the closing of sodium channels and hyperpolarization of the rod cell.
164. C) Extracellularly within the colloid fluid of the follicular lumen | Explanation : Thyroid hormone synthesis is unique because key structural steps occur extracellularly. Follicular cells secrete the large glycoprotein thyroglobulin into the central follicle lumen (colloid). Extracellular thyroid peroxidase then oxidizes iodide and couples it directly to tyrosine residues on thyroglobulin inside the colloid.
165. B) Hyaluronidase | Explanation : The acrosome reaction releases several enzymes. Hyaluronidase degrades hyaluronic acid, which acts as the intercellular cement binding the follicle cells of the corona radiata together. This allows the sperm to reach the inner translucent layer, the zona pellucida, which is then cleaved by the protease acrosin.
166. B) Inside enterocytes, lipids are re-esterified into triglycerides and packaged into massive lipoprotein spheres called chylomicrons, which are too large to pass through blood capillary pores. | Explanation : Free fatty acids and monoglycerides enter enterocytes via passive diffusion, but are reconstructed into triglycerides inside the smooth ER. They are then coated with proteins, cholesterol, and phospholipids to form chylomicrons. Chylomicrons exit via exocytosis but cannot enter blood capillaries due to the restrictive basement membrane; instead, they enter the highly permeable lymphatic capillaries (lacteals).
167. B) Accelerated synthesis of 2,3-bisphosphoglycerate (2,3-BPG), shifting the dissociation curve to the right. | Explanation : Under hypoxic conditions at high altitude, red blood cells increase glycolytic production of 2,3-BPG (also called 2,3-DPG). 2,3-BPG binds to the central cavity of the deoxygenated hemoglobin tetramer, stabilizing its T-state (low affinity state). This shifts the oxygen-hemoglobin dissociation curve to the right, lowering O2 affinity and allowing hemoglobin to release oxygen more easily into peripheral tissues.
168. B) Increased firing of glossopharyngeal and vagal nerve afferents to the medulla, causing increased parasympathetic vagal discharge to slow the heart rate. | Explanation : When blood pressure rises sharply, the walls of the carotid sinuses and aortic arch stretch, increasing the firing frequency of their stretch-sensitive baroreceptors. These nerve impulses travel via the glossopharyngeal (CN IX) and vagal (CN X) nerves to the nucleus tractus solitarius in the medulla. The cardiovascular center responds by increasing vagal parasympathetic output to the SA node (slowing heart rate) and inhibiting sympathetic vasoconstrictor output (dilating vessels).
169. C) It dilates afferent renal arterioles, increases GFR, and directly inhibits the secretion of Renin and Aldosterone, promoting sodium and water excretion. | Explanation : ANP protects the cardiovascular system from volume overload. It relaxes renal afferent arterioles, increasing blood flow into glomeruli and accelerating GFR. Simultaneously, it shuts down the renin-angiotensin aldosterone system (RAAS) by blocking renin release from JG cells and aldosterone release from the adrenal cortex. This blocks sodium reabsorption, causing natriuresis (sodium loss) and diuresis (water loss).
170. B) Hormone → G-protein activation → Stimulation of Adenylate Cyclase → Conversion of ATP to cyclic AMP (cAMP) → Activation of Protein Kinase A | Explanation : Epinephrine acting on beta receptors utilizes the classic cAMP second messenger system. Binding activates a stimulatory G-protein (Gs), which activates the membrane-bound enzyme adenylate cyclase. Adenylate cyclase converts ATP into cyclic AMP (cAMP). cAMP acts as a intracellular second messenger, binding to and activating Protein Kinase A (PKA), which phosphorylates intracellular target proteins to carry out the response.
171. C) Respiratory Acidosis with partial metabolic compensation | Explanation : The blood pH (7.21) is below normal (7.35-7.45), indicating acidosis. The pCO^2 is heavily elevated (68 mmHg), which causes the acidosis, proving the root cause is respiratory retention of carbon dioxide due to lung damage (emphysema). The bicarbonate (HCO^{3-}) is slightly elevated (28 mEq/L), showing that the kidneys have begun retaining bicarbonate to neutralize the acid. This confirms a diagnosis of respiratory acidosis with partial metabolic compensation.
172. B) Fracture of the sella turcica disrupting the supraoptic nuclei of the hypothalamus or posterior pituitary, halting ADH release | Explanation : The symptoms (massive polyuria, dilute urine, lack of urinary glucose) describe Diabetes Insipidus, ruling out Diabetes Mellitus (which would show high urine glucose and high specific gravity). Because the patient had a head injury, the physical trauma likely damaged the hypothalamus or the infundibular stalk of the posterior pituitary where Antidiuretic Hormone (ADH/Vasopressin) is stored and released, leading to Central Diabetes Insipidus.
173. B) It catalyzes the ADP-ribose inactivation of Elongation Factor 2 (EF-2), fully arresting host ribosomal protein synthesis. | Explanation : Diphtheria toxin is an A-B fragment toxin. Once the active 'A' domain enters the cytoplasm of a host cell, it functions as an enzyme that transfers an ADP-ribose group from NAD^+ onto a modified amino acid residue of Elongation Factor 2 (EF-2). This inactivates EF-2, blocking the translocation step of translation on ribosomes, completely halting protein synthesis and killing the cell.
174. B) The Secretory Component derived from the epithelial polymeric immunoglobulin receptor | Explanation : Dimeric IgA produced by plasma cells in the lamina propria binds to the polymeric immunoglobulin receptor (pIgR) on the basolateral surface of mucosal epithelial cells. As the complex is transported through the cell and cleaved at the apical membrane, a fragment of the receptor remains wrapped around the antibody hinge. This is the Secretory Component, which structurally masks cleavage sites and protects IgA from being broken down by harsh digestive enzymes.

175. B) It possesses a partially double-stranded DNA genome that uses a viral reverse transcriptase enzyme to transcribe its genomic DNA from an RNA pre-genome intermediate. | Explanation : Hepatitis B is a DNA virus, but it replicates via an RNA intermediate using reverse transcription. Once inside the host nucleus, its partial DNA is repaired into a circular molecule, which is transcribed into a long 'pre genomic' RNA. This pre-genomic RNA is then packaged into new viral cores in the cytoplasm, where viral reverse transcriptase copies it back into partial viral DNA before export.
176. C) Quantitative Real-Time PCR (RT-PCR) tracking viral HIV-1 RNA copy numbers | Explanation : During the initial 'window period' (the first 2–3 weeks), the human body has not yet produced detectable levels of anti-HIV antibodies, rendering standard antibody ELISA and Western Blot assays completely false-negative. A Nucleic Acid Test (NAT) using quantitative RT-PCR directly amplifies and measures the actual viral RNA circulating in the blood plasma, allowing detection of infection within 10 to 14 days post-exposure.
177. B) Alpha-1-Antitrypsin | Explanation : Tracy the sheep was a famous transgenic mammal engineered by scientists to produce high volumes of human Alpha-1-Antitrypsin (AAT) in her milk. Alpha-1-Antitrypsin is a protective protease inhibitor; individuals with a genetic deficiency in AAT suffer from unchecked elastase activity that destroys lung alveoli, leading to severe hereditary emphysema.
178. B) Pre-existing circulating host antibodies (such as anti-ABO or anti-MHC Class I) binding to graft vascular endothelium, instantly triggering the complement cascade and widespread thrombosis. | Explanation : Hyperacute rejection occurs almost instantly upon restoring blood flow to the transplant. It is caused by pre-existing circulating antibodies in the recipient's blood (due to prior mismatched blood transfusions, multiple pregnancies, or past transplants). These antibodies bind immediately to antigens on the donor organ's vascular endothelial cells, activating the complement system, which leads to endothelial destruction, platelet aggregation, blood clotting, and rapid infarction of the graft.
179. B) Complete anatomical regression of the right ovary and right oviduct, leaving only the left functional side. | Explanation : To optimize aerodynamic efficiency and minimize carrying weight during flight, almost all female birds develop only a single, functional left ovary and left oviduct. The corresponding right structures remain fully vestigial and regress during early development.
180. B) Migratory waterfowl and the last surviving native population of the Wild Water Buffalo (Bubalus arnee). | Explanation : Koshi Tappu Wildlife Reserve, located in eastern Nepal, consists of extensive mudflats, reed beds, and freshwater marshes along the Koshi River. It was designated as Nepal's first Ramsar site in 1987 due to its vital role as a trans-Himalayan flyway wintering station for thousands of migratory waterfowl, and for harboring the last remaining wild population of 'Arna' (Wild Water Buffalo) in Nepal.

MAT

181. C) Both Conclusions I and II follow | Explanation : "Only a few" means some Rivers are Lakes and some are not. However, Rivers can still completely fall inside the 'Oceans' boundary without entirely becoming 'Lakes', making Conclusion I a valid possibility. Since all Lakes are Oceans, and no Ocean is a Sea, no Lake can be a Sea. Therefore, "Some Lakes are not Seas" (in fact, all are not) is logically true. Both follow.
182. C) Both statements I and II together are required to answer the question. | Explanation : From I alone, 'pa' could be 'water', 'is', or 'hot'. From II alone, 'pa' could be 'cold', 'water', or 'flows'. Combining both, the only common word in the statements is 'water' and the only common code is 'pa'. Thus, both statements are required to conclude that 'pa' stands for 'water'.
183. B) Only Assumption II is implicit | Explanation : Assumption I uses the extreme word "only", which makes it invalid (there are other pollution sources). Assumption II is valid because the government targets these specific vehicles to improve air quality, assuming they are a significant contributor.
184. A) Only Conclusion I follows | Explanation : Decoding the symbols: $ means >= (greater or equal). @ means > (strictly greater). # means < (strictly less). Statements become: A>B B>=C, C<D Chain: A>B>=C Therefore, A>C (which means A @ C) is True. Comparing D and B: B>=C and D>C The relationship between B and D cannot be uniquely determined. Conclusion II is False.
185. B) JPRIH | Explanation : The logic follows an alternating substitution of (-1) and (+1) positional shifts in the alphabet. N(-1)=M E(+1)=F P(-1)=0 A(+1)=Z (circular), L(-1)=K. Applying to KOSHI: K(-1)=J, O(+1)=P S(-1)=R, H(+1)=1, I(-1)=H So, JPRIH.
186. C) 5:36 PM Wednesday | Explanation : Time elapsed on the defective clock from 8 AM Mon to 6 PM Wed = 58 hours. For every 24 hours of true time, the defective clock moves 24 hours 10 mins (145/6 hours). So, True Time = 58*(24/(145/6)) = 58*(144/145) = 57.6 hours. 57.6 hours 57 hours and 36 minutes. Adding 57 hrs 36 mins to 8 AM Monday gives 5:36 PM on Wednesday.
187. C) 65 | Explanation : Check the differences between consecutive terms: 15-10=5; 22-15=7; 33-22=11; 46-33=13. The differences are consecutive prime numbers: 5, 7, 11, 13. The next prime number should be 17. 46+17=63 But the series has 65. So, 65 is the wrong term. (63+19=82, which matches).
188. C) 39 | Explanation : The pattern across the rows is that the number in the third column is the arithmetic mean (average) of the numbers in the first two columns. Row 1: (18+24)/2=42/2=21. Row 2: (12+14)/2=26/2=13. Row 3: (27+?)/2=33 -> 27+?=66 -> ?=39.
189. A) Quantity A is greater | Explanation : Vowels in "MEDICAL" are E, I, A (3). Consonants are M, D, C, L (4). Treat the vowels as 1 block. Total entities to arrange = 4 consonants + 1 block = 5 entities. These can be arranged in 5! (120) ways. The 3 vowels inside the block can be arranged in 3! (6) ways. Total ways (Quantity A) = 120 * 6 = 720. Since 720 > 700 Quantity A is greater.

190. A) 300 | Explanation : Since 20% failed in both, the percentage of students passing in at least one subject = 100% - 20% = 80%. Using the formula: n(B U C) = n(B) + n(C) - n(B n C) -> 80% = 60% + 70% - n(B n C) -> n(B n C) = 50%. So, 50% of the total students equal 150. Total students = (150*100)/50 = 300.

191. B) X is the brother of W | Explanation : Decode the chain step-by-step: X @ Y -> X is the son of Y. Y * Z -> Y is the mother of Z. (Therefore, X and Z are siblings, and Y is a female). Z # W -> Z is the brother of W. Since X, Z, and W share the same mother (Y), they are siblings. Because X is a son (male), X is the brother of W. (Option C is incorrect because W's gender is unknown).

192. C) East | Explanation : Calculate the net rotation. Clockwise turns = +90 and +180 = +270. Anticlockwise turns = -135. Net rotation = +270 - 135 = +135 (Clockwise). Starting from North-West, a 135 degree clockwise turn brings him exactly to the East.

193. C) T | Explanation : Let seats be numbered 1 to 6 clockwise. Assume S sits at 1. Since they face the center, Left is clockwise. Second to the left of S is seat 3. So, P is at 3. Q is opposite S (seat 1), so Q is at seat 4. T is immediate right (anticlockwise) of S, so T is at seat 6. Empty seats are 2 and 5. U is not next to Q(4), so U cannot be at 5. Thus, U is at 2. R takes the last seat 5. The final arrangement is S(1), U(2), P(3), Q(4), R(5), T(6). Opposite to P(3) is T(6).

194. A) B | Explanation : From the clues, we can form mathematical inequalities for weights: 1. D > A > C, 2. E > B, 3. C > E. Combining them: D > A > C > E > B. Thus, Car B is the lightest.

195. C) Friday | Explanation : Since 2024 is a leap year, February 29 exists. The time period from February 29, 2024 to February 28, 2025 is exactly a standard non-leap year length of 365 days. 365 days divided by 7 yields 52 weeks and 1 odd day. Adding 1 odd day to Thursday gives Friday.

196. B) | Explanation: Look at the matrix rows and columns. Each row/col has exactly one Circle, one Square, and one Triangle. Each row/col also has exactly one figure with 1 vertical line, one with 2 horizontal lines, and one with 3 diagonal lines. In the third row, we have a Square (3 diagonal) and a Triangle (1 vertical). The missing figure must be a Circle containing 2 horizontal lines.

197. C) | Explanation: By visualizing the folding of the net, alternate squares in a straight line form opposite faces. Thus, 1 is opposite 3; 2 is opposite 4; 6 is opposite 5. Two opposite faces cannot be adjacent on a folded cube. A is wrong (1 and 3 are adjacent). B is wrong (6 and 5 are adjacent). D is wrong (2 and 4 are adjacent). C shows 1, 2, and 5 which are mutually adjacent faces.

198. A

199. C) | Explanation: In the problem figure, the red dot is situated in a region that is common to the Circle and the Square, but strictly OUTSIDE the Triangle. We must find an option where the Circle and Square intersect, and that intersection is not swallowed by the Triangle. In A, the circle and square intersect entirely inside the triangle. In B and D, the circle and square do not overlap without touching the triangle in a conflicting way. In C, the Circle and Square intersect at the top left, completely outside the Triangle's boundary.

200. B) | Explanation: Figure 'X' consists of an angled "Z" shape with an extended bottom tail, intersected by a straight vertical drop from the center. Upon close inspection, Option B contains these exact tracing paths hidden within its added diagonal and structural lines. C has it flipped, and D replaces the vertical line with a horizontal one.